# SPACE POLICY & LAW COURSE OUTLINE

14 TO 16 OCTOBER 2019
CHARLES CLORE HOUSE, RUSSELL SQUARE, LONDON

## SESSIONS

**MONDAY 14 OCTOBER 2019 –** MORNING 09.30 – 12.45; AFTERNOON 13.45 - 17.00

MORNING SESSION:

1. SPACE ENVIRONMENT AND TECHNOLOGY - 09.30 - 11.00 SM & GD

- Delimitation of Space and Airspace
- The Space Environment and Space Technology: Orbits, signals, types of spacecraft
- Discussion and questions

BREAK - 11.00 - 11.15

2. THE POLICY REGIME OF SPACE - 11.15 - 12.45 JC

- Policy Principles of Space, including Peaceful Use, Cooperation and Non-interference
- Policy development, space policy institutions and organisations
- International and National Institutions, including UN COPUOS - nature and role
- Discussion and questions

LUNCH - 12.45 - 13.45

AFTERNOON SESSION:

3. THE LEGAL REGIME - 13.45 - 15.15 SM

- Sources of Space Law including the Outer Space Treaty, other conventions and treaties; a comparison with other legal regimes such as Antarctic, Maritime and Air Law
- International Responsibility and Liability distinguished
- National space laws and policies
- Discussion and questions

BREAK - 15.15 - 15.30

4. THE LAWS GOVERNING SPACE APPLICATIONS - 15.30 - 17.00 SM

- Sovereignty, Property Rights
- Remote Sensing: UN Principles and Disaster Charter
- Scientific experiment and exploration
- Manned space activities: The ISS & its IGA; Chinese Space Station
- Discussion and questions

**TUESDAY 15 OCTOBER 2019 –** MORNING 09.30 – 12.45; AFTERNOON 13.45 - 17.00

MORNING SESSION:

5. COMMERCIAL SPACE ACTIVITIES - 09.30 - 11.00 SA

- Nature and range of commercial space ventures
- Relevance and scope of national and EU laws
- Spaceflight and Spaceports
- Small Satellite Constellations
- Market Access
- Discussion and questions

BREAK - 11.00 - 11.15

6. SATELLITE COMMUNICATIONS - 11.15 - 12.45 EQ (TBC)

- Satellite Telecommunications and Broadcasting: The Role of the ITU
- Orbital Positions, Harmful Interference
- Spectrum management:
  Frequency Efficiency and Market Reach (Harmful interference, jamming, unauthorised signal interception, frequency priority, EU and the ITU)
- Discussion and questions

LUNCH - 12.45 - 13.45

AFTERNOON SESSION:

7. MILITARY AND SECURITY USES OF SPACE - 13.45 - 15.15 MR

- Un-manned space activities (e.g. drone, robotic)
- Military space activities and weaponisation
- Cyber vulnerabilities
- Environmental and humanitarian factors
- Discussion and questions

BREAK - 15.15 - 15.30

8. RISK, LIABILITY & INSURANCE CONSIDERATIONS - 15.30 - 17.00 NH & NS

- Laws Governing Liability (Risk & Fault): The Liability Convention
- Liability for Debris and Obligation not to Create
- Active Debris Removal - Remediation *vs* Mitigation
- Insurance
- Discussion and questions

**WEDNESDAY 16 OCTOBER 2019** - MORNING 09.30 – 12.45; AFTERNOON 13.45 - 17.00

MORNING SESSION:

9. SPACE SECURITY - 09.30 - 11.00 RC

- Space hazards - Man made and natural
- Space Situational Awareness and Traffic Management
- Space Weather, Meteorites, Debris
- Debris Reduction and Elimination Measures
- Responsible Behaviour, Transparency and Confidence Building Measures
- Discussion and questions

BREAK - 11.00 - 11.15

10. CURRENT AND FUTURE DEVELOPMENTS - 11.15 - 12.45 RC & J-JD

- Constellations, Small Satellites
- Focus on Applications
- Extra-terrestrial habitation
- Discussion and questions

LUNCH - 12.45 - 13.45

AFTERNOON SESSION:

11. CASE STUDIES - 13.45 - 15.15 SM

- Recovery and Return
- Debris (liability, sovereignty, space situational awareness and other concepts)
- Mining (Commercial & State participation, property rights, potential pollution from space, appropriation and the issue of territory vs. resources, and other concepts)
- Active Debris Removal

BREAK - 15.15 - 15.30

12. REVIEW & CONCLUSION - 15.30 - 17.00 SM

- Course Summary, discussion and conclusion

Please return questionnaire & CPD certificate requests at the end of this session.

**LECTURERS**

SA - Dr Sam Adlen, Chief Strategy Officer, Satellite Applications Catapult
JC - James Cemmell, Vice President, Government Engagement, Inmarsat; ISPL Faculty
RC - Professor Richard Crowther, Chief Engineer, UK Space Agency; ISPL Adviser
J-JD - Professor Jean-Jacques Dordain, CNES; former Director-General of ESA; ISPL Adviser
GD - Wing Commander (Ret'd) Gerry Doyle, Senior Prog. Manager, Raytheon; ISPL Faculty
NH - Nick Hughes, Partner, HFW; ISPL Faculty
YH - Yvon Henri, OneWeb; Former Chief, Space Services Department, ITU
SM - Professor Sa'id Mosteshar, Barrister; ISPL Director
MR - Air Commodore (Ret'd) Mark Roberts, Defence & Security, RHEA Grp; ISPL Faculty
NS - Neil Stevens, Space Insurance Expert; ISPL Faculty
EQ - Elizabeth Quintana, Satellite Spectrum Management, OFCOM